10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 5.5 — Real World Problems

Unit 5 | Solved Exercise | New Book (PECTAA)

Exercise 5.5 (Solved)

In every question first write the unknown as x, then make the equation from the given condition. Time = distance / speed, and work done in 1 hour = 1 / time. Reject the negative value, because speed and time cannot be negative.

1. A train travels a distance of 240 km at a uniform rate. If it had finished 4 km an hour slower, it would have taken 2 hours more over the journey. Find its rate of travelling.

Let the uniform speed = x km/h

Time taken at x km/h = $\frac{240}{x}$ hours

Time taken at (x − 4) km/h = $\frac{240}{x-4}$ hours

Given: $\frac{240}{x-4} - \frac{240}{x} = 2$

Multiply both sides by x(x − 4)

$240x - 240(x-4) = 2x(x-4)$

$240x - 240x + 960 = 2x^2 - 8x$

$2x^2 - 8x - 960 = 0$

$x^2 - 4x - 480 = 0$ (dividing by 2)

$x^2 - 24x + 20x - 480 = 0$

$x(x-24) + 20(x-24) = 0$

$(x-24)(x+20) = 0$

$x - 24 = 0$ or $x + 20 = 0$

$x = 24$ or $x = -20$

Speed cannot be negative, so x = 24

Answer: Rate of travelling = 24 km/h

2. Arshia and Ibraheem complete a job together in 4 hours. If Arshia takes 6 hours, then find how much time will Ibraheem take?

Let Ibraheem take x hours to finish the job alone

Arshia's work in 1 hour = $\frac{1}{6}$

Ibraheem's work in 1 hour = $\frac{1}{x}$

Work of both in 1 hour = $\frac{1}{4}$

So $\frac{1}{6} + \frac{1}{x} = \frac{1}{4}$

Multiply by 12x: $2x + 12 = 3x$

$12 = 3x - 2x$

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$x = 12$

Answer: Ibraheem will take 12 hours

3. One pipe fills water in 5 hours, another in 8 hours. Second pipe is closed after 2 hours. Find the total time.

Work of first pipe in 1 hour = $\frac{1}{5}$

Work of second pipe in 1 hour = $\frac{1}{8}$

Both pipes remain open for 2 hours

Work of first pipe in 2 hours = $2 \times \frac{1}{5} = \frac{2}{5}$

Work of second pipe in 2 hours = $2 \times \frac{1}{8} = \frac{2}{8} = \frac{1}{4}$

Work done in 2 hours = $\frac{2}{5} + \frac{1}{4}$

$= \frac{8}{20} + \frac{5}{20} = \frac{13}{20}$

Work left = $1 - \frac{13}{20} = \frac{7}{20}$

Time of first pipe to do $\frac{7}{20}$ of work = $\frac{7}{20} \times 5 = \frac{7}{4}$ hours

Total time = $2 + \frac{7}{4}$

$= \frac{15}{4}$ hours = 3 hours 45 minutes

Answer: Total time to fill the tank = 3 hours 45 minutes

4. Huria can complete a project in 12 hours by working alone. If Abdul Hadi joins her and they finish it together in 5 hours, how long would it take Abdul Hadi to do the project alone?

Let Abdul Hadi take x hours alone

Huria's work in 1 hour = $\frac{1}{12}$

Abdul Hadi's work in 1 hour = $\frac{1}{x}$

Together they finish the project in 5 hours

So $\frac{1}{12} + \frac{1}{x} = \frac{1}{5}$

Multiply by 60x: $5x + 60 = 12x$

$60 = 12x - 5x$

$60 = 7x$

$x = \frac{60}{7}$

$x = 8\frac{4}{7}$ hours (about 8 hours 34 minutes)

Answer: Abdul Hadi alone would take 60/7 hours ≈ 8.6 hours

5. Two cars start from opposite towns and head towards each other. The distance between them is 240 km. One travels at x km/h and the other at (x + 10) km/h. They meet after 2 hours. Find the speed of each car.

Distance covered by the first car in 2 hours = 2x km

Distance covered by the second car in 2 hours = 2(x + 10) km

Sum of the distances = 240 km

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$2x + 2(x + 10) = 240$

$2x + 2x + 20 = 240$

$4x + 20 = 240$

$4x = 220$

$x = 55$

Second speed = $x + 10$

$= 55 + 10$

$= 65$ km/h

Answer: Speeds of the cars = 55 km/h and 65 km/h

6. Rashid can paint a house in 6 days, but if he gets a helper he can do it in 4 days. How long would it take the helper to paint the house alone?

Let the helper paint the house alone in x days

Rashid's work in 1 day = $\frac{1}{6}$

Helper's work in 1 day = $\frac{1}{x}$

Work of both in 1 day = $\frac{1}{4}$

So $\frac{1}{6} + \frac{1}{x} = \frac{1}{4}$

Multiply by 12x: $2x + 12 = 3x$

$12 = 3x - 2x$

$x = 12$

Answer: Helper alone will take 12 days

7. Ishmal runs 10 km in 2x/(x + 4) hours. If her average speed is 8 km/h, find the value of x.

Distance = 10 km, time = $\frac{2x}{x+4}$ hours, speed = 8 km/h

Speed = $\frac{\text{Distance}}{\text{Time}}$

$8 = 10 \div \frac{2x}{x+4}$

Multiply both sides by $\frac{2x}{x+4}$

$8 \times \frac{2x}{x+4} = 10$

$16x = 10(x + 4)$

$16x = 10x + 40$

$6x = 40$

$x = \frac{40}{6} = \frac{20}{3}$

Answer: x = 20/3 (≈ 6.67)

8. Fahad runs 600 m at a certain pace, and then doubling his pace, does another 600 m. If he took 2½ minutes to cover the distance 1200 m, find the pace he started at, in metres per second.

Let the starting pace = x m/s, then the new pace = 2x m/s

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2½ minutes = 150 seconds

Time for the first 600 m = $\frac{600}{x}$ seconds

Time for the next 600 m = $\frac{600}{2x}$ seconds

$\frac{600}{x} + \frac{600}{2x} = 150$

Multiply both sides by 2x

$1200 + 600 = 300x$

$1800 = 300x$

$x = 6$

Answer: Starting pace = 6 m/s

9. A cyclist travels 30 km at a certain speed. If the speed had been 5 km/h faster, the journey would have taken 1 hour less. Find the speed.

Let the speed = x km/h, then the faster speed = (x + 5) km/h

Time at x km/h = $\frac{30}{x}$ hours

Time at (x + 5) km/h = $\frac{30}{x+5}$ hours

Given: $\frac{30}{x} - \frac{30}{x+5} = 1$

Multiply both sides by x(x + 5)

$30(x + 5) - 30x = x(x + 5)$

$30x + 150 - 30x = x^2 + 5x$

$150 = x^2 + 5x$

$x^2 + 5x - 150 = 0$

$x^2 + 15x - 10x - 150 = 0$

$x(x + 15) - 10(x + 15) = 0$

$(x + 15)(x - 10) = 0$

$x = -15$ or $x = 10$

Speed cannot be negative, so x = 10

Answer: Speed of the cyclist = 10 km/h

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